Shiham howdhury

Shiham | Shiham Chowdhury | shiham@umich.edu | Portfolio

DUCATION

University of Michigan — **August 2022 May 2026**
Bachelor of Science in Electrical Engineering — *Ann Arbor MI*

Technical Skills

Languages: Java, Javascript, C++, Python, MATLAB
Developer Tools: Visual Studio, Jupyter Notebook, LTSpice, React, Git
Hardware: Soldering, Oscilloscope, PCB Design, Network Analyzer, Arduino, Raspberry Pi

Projects

Environment Monitoring Device *Arduino IDE, Amazon Web Services AWS)* — Link to Project

Designed an IoT-based system using the ESP32 and DHT22 sensor for accurate, real-time climate data collection.
Integrated AWS IoT Core for secure data transmission and used AWS S3 buckets for scalable and efficient storage.
Developed a React front-end application that displays data from S3 using EC2 MATLAB scripts.

Word Occurrence Prediction Algorithm *C++* — Link to Project

Built a C++ algorithm that learns inputted language to predict the natural occurrence of a word.
Utilized machine learning algorithms to analyze and categorize words into pair types for improved prediction accuracy.
Employed recursive binary search trees to efficiently parse and manage learned data, facilitating quick computations of log-probability scores.

Euchre Game *C++* — Link to Project

Designed and implemented a fully functional Euchre game in C++ that supports up to 4 players.
Integrated comprehensive game logic to enforce Euchre rules and maintain accurate game states.
Implemented strategic decision-making algorithms based on game state to simulate CPU opponents.

Temperature Controlled Fan *Arduino IDE, Arduino Uno* — Link to Project

Designed and built an Arduino-based device to control a fan's operation based on temperature readings and proximity detection.
Utilized a DHT11 sensor to monitor temperature and humidity, with data displayed on a 16x2 LCD screen for real-time feedback.
Implemented logic to activate DC motor-powered fan when temperature is high enough and deactivate when temperature is low or when ultrasonic sensor detects an object getting too close.

XPERIENCE

eTitle IT Development — May 2025 - August 2025

Helped development a web-based foreclosure management system designed for user convenience
Tested updated merge templates using sample foreclosure cases to ensure proper functionality
Optimized merge template scripts to remove hard-coded elements and replace outdated data

Wolverine Solar Car Team — August 2023 - present

Divided battery cells into multiple modules with thermistors to monitor temperature and implemented a MOSFET to switch cooling fans on when temperatures reach 40°C.
Programmed a PID control script on an STM32 microcontroller to dynamically adjust cooling fan speeds according to battery temperature, adjusting gains to account for natural error.
Designed a low-pass lter using MATLAB to eliminate high-frequency noise and short-term fluctuations caused by electrical interference and transient spikes.